

DeckForge AI — speaker notes

Bachelor defence · Kylian Algoet · 2026-09-02

Deck	15 slides (13 core, 2 supporting)
Slide narration	14:56 at 130 wpm, including the spoken demo handoff
Live demo	4:00 target, timed separately — docs/presentation/demo-script.md
Combined	18:56 against a 20:00 slot — buffer 1:04
Backup plan	docs/presentation/demo-backup-plan.md — read it before presenting
Jury questions	docs/presentation/jury-questions.md
Built from	commit 6ca01a6 on 2026-08-14

Every time above is an ESTIMATE from the word count of these notes — not a measured rehearsal. No rehearsal has been run. The buffer is 1:04, so the estimate does not have to be wrong by much to matter: **rehearse against a clock before the defence** and re-cut from the notes if it runs long. Do not solve an overrun by speaking faster, and do not drop a bounded claim to save time — the concessions on the limits, reproducibility, base-model and LoRA slides are enforced by validate_slides.py and are not negotiable against the clock.

CORE · title · ~15 s spoken (33 words)

ON THE SLIDE

Kylian Algoet · Multimedia & Creative Technologies
Erasmushogeschool Brussel · Bachelor defence, 2 September 2026

SAY

Good morning. I am Kylian Algoet, and this is DeckForge AI. What I am defending is not the demo. It is the research: what fits in eight gigabytes, and how I established it.

Next: The assignment, and the question under it

CORE · statement · ~70 s spoken (152 words)

ON THE SLIDE

A skateboard manufacturer wants **customer-designed decks** — without a designer in the loop for every order.

How can a **locally fine-tuned** diffusion model, conditioned on **text and a reference image**, generate decal artwork in **multiple visually distinct styles** with **reproducible quality** on **8 GB of VRAM**?

docs/00-project-brief.md · thirteen mandatory requirements · RQ1–RQ12 · assessed on the research process, not the artefact

SAY

A skateboard manufacturer wants customers to design their own decks. Today that needs a designer for every order, which does not scale to one-offs.

The assignment sets four things: the model must be self-trained, it must take both a prompt and a reference image, the result must land on an interactive three-D deck, and it must be reproducible by someone else. The hard part is none of those individually — it is that all of it runs on one consumer laptop GPU.

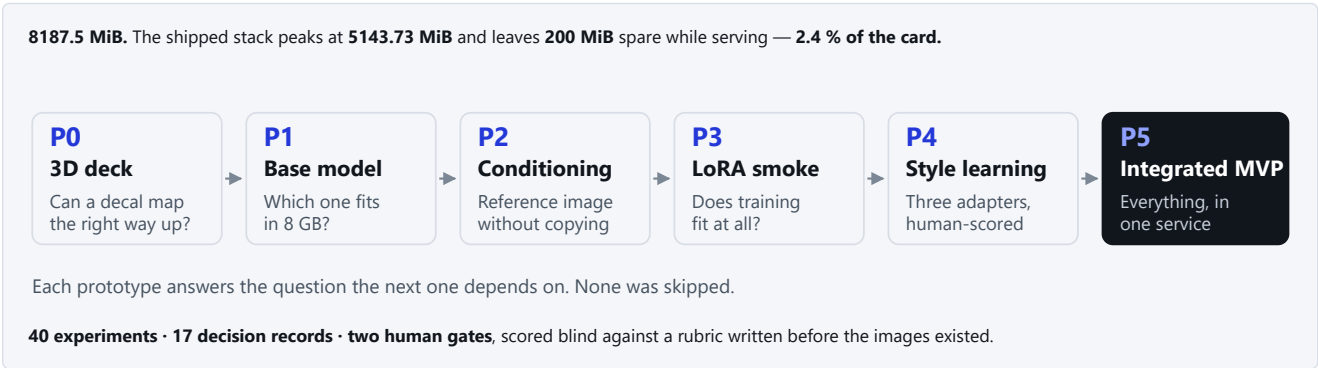
Every clause of the question was chosen to be falsifiable. Distinct styles is a claim needing evidence, not an adjective. Reproducible quality is where my answer splits in half. And eight gigabytes is a number, so a configuration either fits or it does not.

This is assessed on the research process. So when I show you a failure later, I am not apologising — I am showing you the deliverable.

Next: One constraint, and how the work was done

CORE · figure · ~71 s spoken (154 words)

ON THE SLIDE



SAY

The number the whole project is built around. Eight gigabytes — the figure the driver reports, not the marketing one. The production stack peaks just over five gigabytes and leaves about two hundred megabytes spare while actually serving. Two point four per cent. It fits, and it does not fit easily. I never call it headroom.

Six prototypes, each answering the question the next depends on. P0 is the one worth defending. Building the three-D deck before I had any artwork looks like the wrong order. It was not: it made the deck's aspect ratio a hard input to the image problem, and it caught an orientation fault that conveniently square test images would have hidden until the end.

Every choice ran the same loop — alternatives, criteria fixed before the experiment, then the measured result. Several decision records reject the option I expected to pick, and the next two are both like that.

Next: The dataset was built, not downloaded

CORE · split · ~58 s spoken (125 words)

ON THE SLIDE

- **148 items**, three styles, built for this project — holdout of 7 never seen in training
- Every item is **CC0, public domain, or made for this project**. No third-party licensed work
- **Licence and source URL recorded before the item entered a split** — a gate, not an afterwards audit
- Captions are **style-only**. That was tested against verbatim descriptions, and style-only won

SAY

The dataset is mine, and it is where a lot of the work went. A hundred and forty-eight items: ukiyo-e woodblock prints from the Metropolitan Museum, minimal-geometric that I generated and curated, and retro silkscreen posters from the Library of Congress. The seven-item holdout never entered a training run, so I had something honest to test memorisation against.

Licensing was decided before training rather than justified afterwards, and the ordering is the point. Everything is CC-zero, public domain, or made by me, so the fair-use argument never has to be made. A licence went into the manifest as a condition of entering a split.

The captions were an experiment, not a preference: verbatim descriptions taught the model to reproduce the caption rather than the style.

Next: SDXL was better, and was rejected

CORE · split · ~82 s spoken (178 words)

ON THE SLIDE

- SDXL scored **better** than SD 1.5 on my own rubric, at its native resolution
- Rejected anyway: **10 738 MiB allocated** at 1024², on a 8187.5 MiB card
- Windows **spilled into host memory instead of failing**. All thirty runs reported success
- SD 1.5 selected — not because it wins, but because **it is the one that fits at a useful resolution**

SAY

The decision I expected to go the other way, and the one that set the method for everything after.

I benchmarked SD 1.5 against SDXL on identical prompts and seeds. At its designed resolution — these images — SDXL is better. My own rubric scores say so, and I left that in the report rather than quietly dropping the comparison.

I rejected it anyway: ten point seven gigabytes allocated at one thousand and twenty-four square, on an eight gigabyte card.

The important part is that it did not crash. Windows spills GPU memory into system RAM, so all thirty runs completed and reported success. Judged by exit code, SDXL was fine — and I would have built the whole project on it and found out in the last month.

That is where the habit came from that I would most want to keep. A successful run is not a run that fitted. Those are different claims, and modern systems blur them. So every memory figure after this one is read against the ceiling, never against whether anything crashed.

Next: The free method lost

CORE · split · ~69 s spoken (150 words)

ON THE SLIDE

- Two ways in for a reference image: **img2img** or **IP-Adapter**
- img2img costs **zero extra VRAM** — decisive on this budget, and it lost anyway
- At the deck format it **returned its own input**. Every near-copy flag came from img2img
- **IP-Adapter selected**, scale 0.55. *A method that reproduces its input is not a generator.*

SAY

The decision where the criteria mattered most.

Two realistic ways to condition on a reference. img2img starts the diffusion from the reference itself and costs nothing extra. IP-Adapter encodes it and injects it through cross-attention, which costs VRAM I did not have.

On a card with two hundred megabytes spare, costs extra VRAM should have ended the argument.

Look at the pairs. Left is the reference the user supplied, right is what img2img returned at the deck format. Several are perceptually the same image. Every near-copy flag in that milestone came from img2img; none came from IP-Adapter.

So I rejected the cheaper method on what it does, not what it costs. A system that hands back the customer's own upload with a filter on it is not a generator — and on a product where people upload artwork they care about, that is also the failure mode with legal consequences.

Next: Training longer made it less obedient

CORE · two-figures · ~87 s spoken (189 words)

ON THE SLIDE

Longer training bought style and spent obedience. Three per-style LoRA adapters, rank 8, applied at 0.7 — two ship at step 300, ukiyo-e at 600. **The claim is feasibility, not superiority.**

SAY

My favourite result, and I only have it because of a process decision.

Same style, same prompts, same seeds. Left is three hundred training steps, right is six hundred. Six hundred is more stylised — and less obedient. Prompt adherence dropped from four to three while style consistency held at five. The model got better at the style by getting worse at listening to you. On a product where the customer types what they want, that is the wrong trade.

So two of three adapters ship from step three hundred, not the six hundred they trained to. I only saw that because I checkpointed four times along each run and had them scored blind. One global step count would have got two styles out of three wrong.

On LoRA itself: full fine-tuning does not fit, and from scratch needs a cluster. LoRA freezes the base model and trains a small pair of matrices beside it.

The honest bound, before you ask: I did not prove LoRA beats the alternatives. Textual Inversion, DreamBooth and full fine-tuning were screened on criteria and never run, so that question is bounded, not answered.

Next: What did not work

CORE · split · ~84 s spoken (183 words)

ON THE SLIDE

- **retro-poster ships as a PARTIAL PASS** — it learned the posters' frames and lettering. Labelled in the interface; not dropped, not re-scored until it passed
- **How many images does a style need? INCONCLUSIVE** — 12, 24 and 44, non-monotonic, one run each
- **No second independent human rater** — one human approver plus AI-assisted visual analysis, so no inter-rater agreement
- SD 2.1 was authentication-gated; memorisation is **bounded, not settled**

SAY

A limitations slide that is only a formality is worse than none.

One of my three styles is a partial pass. The training material is silkscreen posters, so almost all of it has a frame and display lettering as part of the composition, and the adapter learned all of that. Ask for a mountain and you get a mountain inside a poster frame. I could have dropped the style, or re-scored it until it passed. I shipped it labelled.

Second, a question I set out to answer and did not. Same style on twelve, twenty-four and forty-four images, everything else fixed. I expected a plateau; what I got was non-monotonic, one run per condition, so I cannot separate a real effect from variance. It would have been easy to write that about forty images is enough, and nobody would have checked.

Third, and I would rather raise it myself. Scoring was AI-assisted: ChatGPT proposed visual analysis and scores at the gates, and I reviewed and approved every recorded score. But there was no second independent human rater, so I cannot report inter-rater agreement.

Next: The integrated system

CORE · figure · ~58 s spoken (125 words)

ON THE SLIDE

Generation goes **direct to 512×1536** — not square-then-crop. **One worker is asserted in code**, because a second resident pipeline does not fit: a concurrent request gets a clean refusal, never an out-of-memory crash.

Browser

React 19 + Vite
Three.js / R3F deck
Procedural geometry,
runtime texture swap

```
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<text x="428" y="148" font-size="21" fill="#46525f">Busy lock &#8594; 409, never OOM</text>
<text x="428" y="180" font-size="21" fill="#46525f">Upload validation, timeouts</text>
<text x="428" y="206" font-size="21" fill="#46525f">SHA-256 checkpoint verify</text>

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<text x="858" y="116" font-size="21" fill="#c8d0dc">Stable Diffusion 1.5</text>
<text x="858" y="148" font-size="21" fill="#c8d0dc">+ one per-style LoRA</text>
<text x="858" y="180" font-size="21" fill="#c8d0dc">+ IP-Adapter (optional)</text>
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<text x="342" y="126" font-size="18" fill="#7b8794">HTTP</text>

<text x="4" y="296" font-size="22" fill="#46525f">Two processes, one machine. Single-worker is a correctness requirement enforced in
```

SAY

Two processes on one machine: a React frontend with the Three.js deck, and a FastAPI service that owns the model.

The single worker is what I would point at. It is not a default I never changed — it is asserted at startup, because a second resident pipeline does not fit in what is left. Two simultaneous requests, and the second gets a clean refusal rather than an out-of-memory crash that takes the first down with it. Refusing correctly is a feature here.

Generation goes straight to the deck's tall ratio rather than generating a square and cropping. I expected that to hurt, because diffusion models are trained mostly on square images. I wrote it down as a hypothesis, tested it, and it was refuted.

Next: Testing, and what it cannot tell you

SUPPORTING · bullets · ~56 s spoken (122 words)

ON THE SLIDE

- **473** product and research tests + **16** report-validation tests = **489 pytest** · 183 vitest · 38 end-to-end
- The **clean-clone test** is the one that counts: empty directory, install from scratch, working system
- It caught an **integrity control that had only ever passed on the machine that wrote it**
- **A green suite does not prove the generator makes good pictures.** Only the human gates did that

[docs/evidence/M8/clean-clone/](#) · report §15

SAY

The number is split on purpose: most of those tests cover the product and the research code, and sixteen more test the report itself — that every quantitative claim still matches the evidence file it came from.

The one that earned its place is the clean-clone test. Clone into an empty directory, install from scratch, reach a working system. It caught a real defect: a checkpoint integrity control that passed every time locally, because the path it checked only existed on my machine. It had been green for weeks while verifying nothing.

The honest limit is the last line. Those tests tell you the code does what I told it to. Not one of them tells you a generated deck looks good.

Next: Reproducible has halves

CORE · split · ~66 s spoken (142 words)

ON THE SLIDE

- **Inference is deterministic.** A clean clone reproduced an earlier output **byte for byte**, three days later, in a fresh environment
- **Training is not.** Adapter initialisation draws from an **unseeded global generator** — a run cannot be repeated from its recorded seed
- So the checkpoints are **artifacts, verified by hash on every request** — not recipes
- A defect that produces perfectly valid-looking runs, and is **invisible unless you compare weights**

SAY

The primary question asks for reproducible quality, and this is where my answer splits in half.

Inference is deterministic — and not just on my machine on the same day. A clean clone in a fresh environment, three days later, reproduced an earlier output byte for byte. Same hash.

Training is not reproducible at all from its recorded seed. Two runs with identical configuration and an identical seed produced different weights, because the adapter initialisation draws from an unseeded global generator. The seed I was diligently recording never governed the part that mattered.

So the checkpoints are artifacts, not recipes. Their hashes are verified on every request, and the deployment docs say plainly that three required files cannot be regenerated.

The part worth pushing on is that this defect produces runs that look completely fine. It is invisible unless you compare weights.

Next: Local, and chosen

SUPPORTING · bullets · ~63 s spoken (137 words)

ON THE SLIDE

- Four options compared on nine criteria. **Selected: native local run + pre-generated backup assets**
- **200 MiB spare prevents a second resident pipeline**, so multi-worker inference is unsupported
- Docker GPU passthrough was **never verified on this machine**; unmeasured overhead is disqualifying on this margin
- **Public cloud was deliberately not selected** — cost, a demo-day network dependency, and no research gain
- Reproducibility rests on the **runbook and the clean-clone proof**, not on a public URL

DR-014 · alternatives A–D · EXP-034

SAY

Is it actually deployed is a fair question, and the answer needs no spin. There is no public URL. It runs locally, and that was a choice between four compared options rather than something I ran out of time for.

Not cloud, because it adds a network dependency on the one day everything must work and — the deciding part — answers nothing. The question is what fits in eight gigabytes of consumer hardware. A rented data-centre card replaces that question rather than answering it.

Not Docker, because GPU passthrough was never verified here and its overhead is unmeasured. On this margin, that is disqualifying.

What I do claim is reproducibility, and it rests on the runbook and the clean-clone test — a stronger claim than a URL, which only proves it runs where I put it.

Next: Conclusions

CORE · statement · ~56 s spoken (122 words)

ON THE SLIDE

Yes — and the configuration that fits is specific and measured. SD 1.5 at 512×1536, one per-style LoRA at 0.7, IP-Adapter at 0.55.

Bounded three ways: **the margin is 2.4 % · reproducible describes inference, not training · three styles, and one is partial.**

Not concluded: that LoRA beats the alternatives · that SD 1.5 is the best base model · that any image count is a threshold · that a green suite means good pictures.

SAY

The answer is yes, and the useful part is the specificity: I can name the configuration that fits, with the memory reading behind each element.

But the bounds matter more than the yes, so I state all three every time. The margin is two point four per cent — it fits, not comfortably. Reproducible describes inference, not training. And multiple distinct styles means three, one of which is partial.

The second half is what I would most like to be judged on. Those are things I could easily have concluded and did not, because I did not measure them. Each of them is one easy sentence, and none is supported by what I measured. So none of them is in the report.

Next: Live demo

CORE · demo · ~115 s spoken (250 words)

ON THE SLIDE

Prompt → style → **512×1536 decal** → 3D deck. Running on this laptop's GPU. Nothing leaves the machine.

If the GPU does not cooperate, that is a live-demo problem and not a project problem — there is a recorded run ready and I will say which one you are looking at.

SAY

Spoken handoff, about fifteen seconds, while moving to the browser:

"This runs locally, on this laptop's GPU. The three styles are not prompt tricks — they are adapters I trained on the dataset I showed you."

Then hand off to the timed demo script. Four minutes. Do not improvise it.

Browser should already be at the printed URL, one tab, no console, notifications off.

Press Generate early and keep talking. A warm generation is about thirteen seconds, a cold one about thirty. Both are dead air if you stop, and both are enough time to explain the progress panel if you do not. Suggested prompt: *a mountain and a rising sun*, style **Ukiyo-e** — the unambiguous pass, and it reads well at a distance.

Do not generate with retro-poster. It is a partial pass; showcase it in words and generate with a full pass.

Then: the reproducibility metadata, the 3D deck with its stretch disclosure, and `Upload your own decal` — which is also the fallback, so the room has already seen it work.

If it fails: say the line on the slide, then go to `Upload your own decal` with a pre-generated PNG. Do not reload, do not retry, do not open a terminal, do not run a second generation. The fallback ladder has four rungs and every one of them is under thirty seconds.

If the output ignores half the prompt — **that is not a failure, it is the limitations slide**. Say so and use it.

Next: What I would change

CORE · bullets · ~46 s spoken (99 words)

ON THE SLIDE

- **Repeat every condition before drawing a curve.** The image-count question is inconclusive because I ran each condition once — a planning mistake, not a resource one
- **Seed everything, then verify by comparing weights** rather than trusting the log
- **Get a second rater.** One approver cannot produce inter-rater agreement, and I could have arranged one
- **40 experiments · 17 decision records · six prototypes** — and conclusions that stop where the evidence stops

github.com/KylianAlgoet/Selftrained-and-deployed-AI-image-generator · docs/learning-outcome-traceability.md

SAY

Three things I would change.

Repeat every condition before drawing a curve. The image-count question is inconclusive because I ran each condition once — a planning mistake, not a resource one.

Seed everything, then verify by comparing weights rather than trusting the log. I was confident about reproducibility for weeks, and only found out because I tried to reproduce a result instead of assuming I could.

And get a second rater.

That last line is what I would like to be judged on: conclusions that stop where the evidence stops. Thank you — I am happy to take questions.

Next: — end of deck —